and the initiation factors (IF1 and IF2 and GDP) are released (**Fig. 5.13 b**).

Elongation is the next phase of translation. Once both subunits of the ribosomes are assembled with the mRNA, binding sites for two charged tRNA molecules are formed. The sites in the ribosome are referred to as the aminoacyl site (A site), the peptidyl site (P site) and the exit site (E site). The charged initiator tRNA binds to the P site. The next step in prokaryotic translation is to position the second tRNA at the 'A' site of the ribosome to form hydrogen bonds between its anticodon and the second codon on the mRNA (step1). This step requires the correct transfer RNA, another GTP and two proteins called elongation factors (EF-Ts and EF-Tu).

Once the charged tRNA molecule is positioned at the A site, the enzyme peptidyl transferase catalyses the formation of peptide bonds that link the two amino acids together (step 2). At the same time, the covalent bond between the amino acid and tRNA occupying the P site is hydrolyzed (broken). The product of this reaction is a dipeptide which is attached to the 3' end of tRNA still residing in the A site. For elongation to be repeated, the tRNA attached to the P site, which is now uncharged is released from the large subunit. The uncharged tRNA moves through the **'E' site** on the ribosome.

The entire mRNA-tRNA-aa1-aa2 complex shifts in the direction of the **'P' site** by a distance

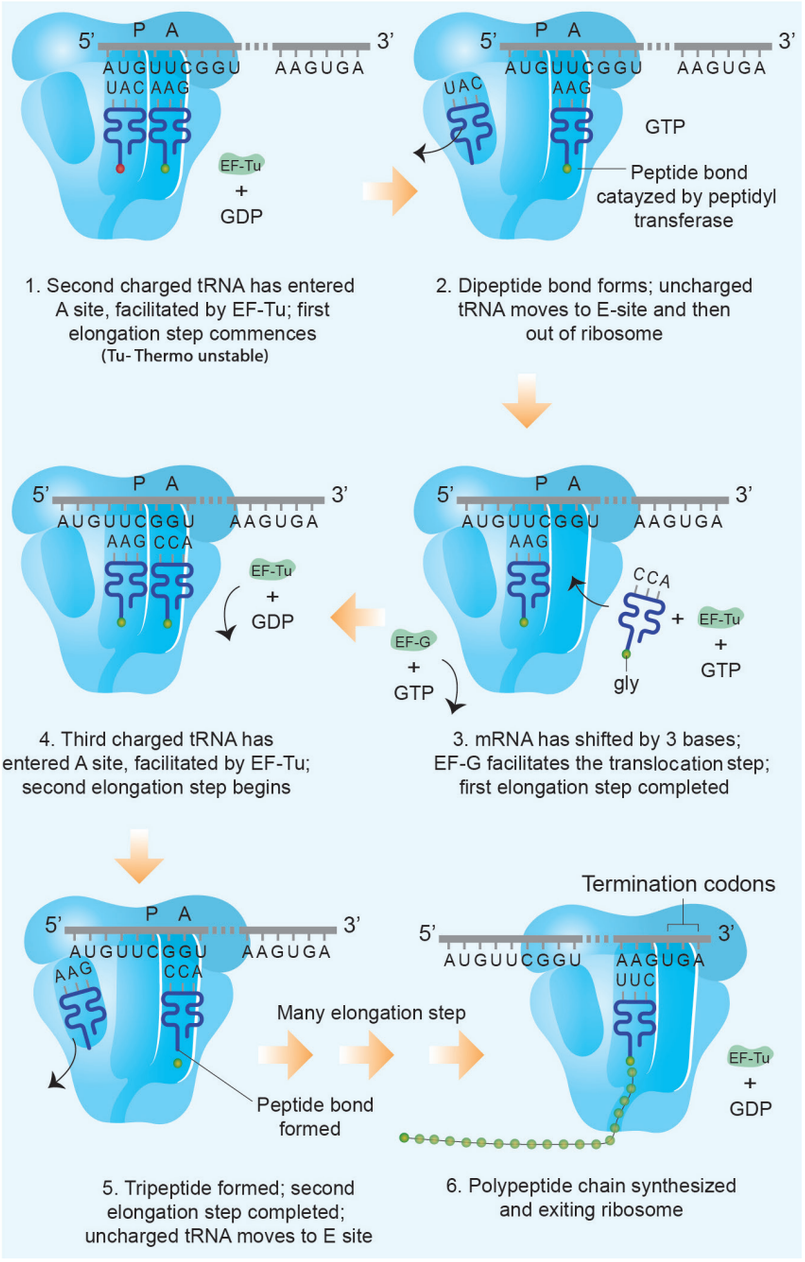


Fig. 5.13 c- Elongation of the growing polypeptide chain during translation

of three nucleotides (step 3). This step requires several elongation factors (EFs) and the energy derived from hydrolysis of GTP. This results in the third triplet of mRNA to accept another charged tRNA into the A site (step 4). The sequence of elongation is repeated over and over (step 5 and step 6). An additional amino acid is added to the growing polypeptide, each time mRNA advances through the ribosome. Once a polypeptide chain is assembled, it emerges out from the base of the large subunit (**Fig. 5.13 c**).

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